

DRISHYAM

Build Roadmap

SIH26189 — AI-Powered Criminal Network Analysis System
Ministry of Home Affairs · NCRB Women Safety Division

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This is the work agreed after a walkthrough of the running product as an investigator would use it. Everything here is additive: the provenance chain, the network layer, the source viewer, the four report profiles and the benchmark already exist and are not revisited. What follows is what an investigator still cannot do, what they cannot understand, and what the system claims but does not yet prove.

Where this stands

Phases 0 to 9 of the original change roadmap are complete: the criminal-network vocabulary, entity-to-entity relationships, network analytics, source adapters, temporal rules, image preprocessing, the workspace surface, the case assistant and the benchmark. Since then the report was rewritten around the network, split into four audience profiles, given numbered findings and source crops, and a §63 certificate form. The landing page was rebuilt to argue the criminal-network case. 372 backend tests pass.

Three findings from the walkthrough drive most of this roadmap:

#	Finding	Where
1	The stated role — Complainant, Accused, Victim, Witness, Driver, Owner — is matched by the person extractor and then thrown away. Only the name is kept. The first question any investigator asks is the one piece of information the system deliberately discards.	patterns.find_person_names()
2	There is no entity endpoint at all. An investigator cannot click a name and see who it is, where it came from, or whether it is known to another case. The data exists; nothing surfaces it.	no /entities/{id}
3	The audit hash chain is written on every action but never verified. trustify/verify checks the report and manifest hashes — which works — and only echoes the chain head. A hash chain nobody walks is not tamper-evidence; it is a table that looks like one.	trustify.verify_receipt()

Phase A — The investigator can understand what they are looking at

The product is accurate and hard to read. These four items change nothing about what it finds and everything about whether a working officer can act on it.

ID	Work	Effort	Depends on	What it solves
A1	Capture the stated role. Keep the label the person pattern already matches, and store it on the entity as an observation with its own source.	0.5 day	—	Unlocks every summary below. Without it the system cannot say whether a person is the complainant or the accused.
A2	Entity summary card. Click a node, get four sentences: role as stated, where it was found, what it connects to, what it does not connect to. Template-driven from stored facts.	1 day	A1	The investigator's first question, answered where they ask it. Every clause is citable.
A3	Confidence as a sentence. Replace 0.92 with “stated directly in a call record”. Keep the number available on hover for the record.	0.5 day	—	A number no officer can calibrate becomes a fact they can weigh.
A4	Every page opens with the work. One line at the top of each view saying what to do, derived from the data already on it.	1 day	—	Turns a set of dashboards into a set of instructions.

Phase B — The questions an investigator actually asks

Three gaps that stop the product being usable on a real case rather than a demo one.

ID	Work	Effort	Depends on	What it solves
B1	Entity profile page. Aliases and how each was written, every occurrence with its source, every relationship, that entity's own timeline, and its appearances in other cases.	2 days	A1	“Who is Ravi Kumar and what is his record.” The single largest missing surface.
B2	Incident date at case creation. One field on the case form.	2 hours	—	The entire temporal layer — contact before, during and after the incident, communication bursts — is built and dormant because no case declares a window.
B3	Cross-case identity check. “This number appears in two other cases; here is the officer to contact,” without disclosing the other case's content.	2 days	E3	Siloed districts are the problem MHA actually has. Cross-case links today are case-to-case, not identity-to-identity.

Phase C — Alerts that tell a story

The alert layer states facts correctly and reads like a log. An investigator needs the sequence. It must remain a sequence: the system states what happened and where it is written, and never why.

ID	Work	Effort	Depends on	What it solves
C1	Narrative pattern alerts. Order the sourced facts by time and render them as a sequence, with each line carrying its file and place, and a closing line stating that the meaning is the reader's.	2 days	—	Reads as a story, cites every clause, asserts no intent.

ID	Work	Effort	Depends on	What it solves
C2	Five new pattern rules: relay (A→B→C inside a window), converging location, late-arriving identity, vehicle corridor, and sudden silence between a pair that had been in regular contact.	2 days	—	All computable from data already stored. Silence in particular is a strong signal nothing currently reports.
C3	Alerts page states the work. “Four patterns. Two cluster around one evening. Verify these two links first — if either is wrong the other three fall.”	0.5 day	C1	Priority instead of a list.

Phase D — The report as a working surface

The report is generated, downloaded and then leaves the system. It should be readable, searchable and answerable inside the product first.

ID	Work	Effort	Depends on	What it solves
D1	In-app report viewer. Render pages, search inside them, highlight matches at their exact rectangles. PyMuPDF already provides both.	2 days	—	Review before download, and find a line in a nineteen-page document.
D2	F-xx opens its evidence. Findings are already numbered and carry their file and exact place; wire each to the existing source panel.	0.5 day	D1	“Which evidence was this from” becomes a click rather than a question.
D3	Report Q&A through the offline assistant. Questions of fact answered exactly from case rows. Questions of wording surface the explanation the report already stores.	1 day	D1	Answers without putting report content — victim names, numbers — into a hosted model.

Phase E — Integrity, and where blockchain genuinely belongs

A hash chain already exists and is real. What is missing is proof that it holds, and the one situation where a shared ledger is the right tool rather than decoration: two districts that cannot show each other their data.

ID	Work	Effort	Depends on	What it solves
E1	Chain verification. Walk the audit chain from the first entry, recompute every hash, and report “142 entries, chain intact” or the exact entry where it breaks. Surface as a badge and a report line.	1 day	—	Converts a claim into a demonstration. Today there is no way to answer “prove the chain holds”.
E2	Merkle root per case. A root over the evidence hashes, printed in the report, with an inclusion proof for any single file.	1 day	E1	Proves a file was in the case at report time without disclosing the other files.
E3	Shared hash ledger for cross-case matching. Districts publish sha256(identifier) with a case reference — never the identifier, never case content. A hash discloses nothing and cannot be reversed.	3 days	E1	The one place in this product where no party trusts another, which is the only place a distributed ledger earns its cost.

Phase F — Fewer places to look

The workspace has seventeen views. No investigator holds seventeen locations in their head, and several overlap. Nothing is removed — the same features are reached in fewer places.

ID	Work	Effort	Depends on	What it solves
F1	Seventeen views to eight. Entities & Graph + Network Intelligence → Network. Corroboration + Contradictions + Review Queue → Review. Integrity + Custody/Audit + Processing → Integrity. Reports + Report Versions → Reports.	2 days	—	A navigable product, and a demo that can show every surface inside the time allowed.

Phase G — Finishing the workflow

The product analyses a case well and does not yet help an officer work one.

ID	Work	Effort	Depends on	What it solves
G1	Notes on entities and relations.	1 day	B1	An investigator cannot currently write anything down inside the system.
G2	Export. CSV and JSON for entities, relations and findings.	1 day	—	Nothing exports today; the next system has to be typed into by hand.
G3	Requisition draft. Which numbers, which date ranges, drafted from the network.	1 day	C2	The bridge from analysis to the next investigative action.
G4	What changed since you were last here.	1 day	—	An investigator opening a case should not have to work out what is new.

Order of work

Ordered by what unlocks the most and what an evaluator sees first, not by phase letter. A1 is first because four other items are blocked behind it and it takes half a day.

Step	Items	Effort	Why here
1	A1 + A2 — role capture and the summary card	1.5 days	The investigator's first question, and the cheapest large win. Everything in Phase B reads better afterwards.
2	B1 — entity profile page	2 days	The largest missing surface in the product.
3	E1 — chain verification	1 day	The theme is Blockchain & Cybersecurity and the chain is currently unproven. One day closes that.
4	B2 — incident date	2 hours	One field wakes an entire dormant feature.
5	C1 + C2 — narrative alerts and the new rules	4 days	The largest readability gain after A2.
6	D1 + D2 — report viewer and F-xx links	2.5 days	Closes the loop from report back to evidence.
7	F1 — seventeen views to eight	2 days	Worth as much to the demo as any new feature.
8	E3 — shared hash ledger	3 days	The blockchain story, told where it is true.
9	A3, A4, C3, D3, E2, G1–G4	7 days	Polish, in whatever order time allows.

Roughly 25 working days for one person. Phases C, D and F are independent of each other and of Phase B, so three people can run them in parallel after step 2.

What is deliberately not being built

Each of these was considered and rejected. Recording the reasons matters as much as recording the work, because every one of them will be suggested again.

Not building	Reason
A language model summarising case data	It would send names, numbers and case context to a hosted provider and destroy the strongest claim this product has. The summary does not need one: the facts are structured, and a template-built summary is citable line by line where a generated one is not.
“Why did this person go there”	That is intent. The system states what a source states and refuses to infer meaning. A confident wrong lead costs an investigation more than a missing one, and this restraint is what makes the output defensible.
A distributed ledger with consensus for the case record	Consensus solves disagreement between parties who do not trust each other. A police evidence system has one responsible custodian. A chain written and read by one machine is a database table called a blockchain, and evaluators recognise it. The hash chain is the part that is genuinely needed, and it already exists.
Neo4j, PaddleOCR, or new infrastructure	Binding non-goals from the original roadmap. NetworkX and the current stack carry the analytics; new infrastructure costs schedule and buys nothing before the demo.
Face recognition or biometric matching	Not asked for by the problem statement, and an ethical surface this project should not open.

Standing items

Item	Status
Fifty-plus commits exist only on one laptop. The whole of phases 0–9 and everything since is unpushed. This is the largest single risk to the submission and takes one command to remove.	Open — needs a decision to push

Item	Status
Demo rehearsal. The benchmark case, uploaded to a clean clone, walked end to end three times: network → node → source panel → assistant → four reports.	Open
Case type on the demo case still reads “Digital Payment Fraud”. Create the demo case with the correct type and the correct source categories per file.	Open — five minutes

Nothing in this document claims a capability is implemented unless it was verified in the running system during the walkthrough it came from.